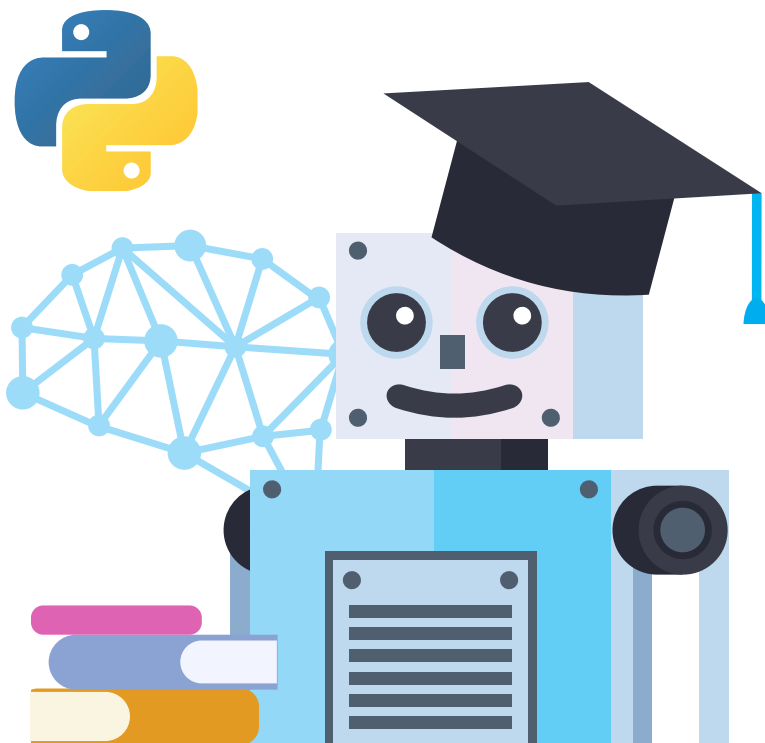


AI: A Few More Useful Python Libraries

In the previous article in this series, we discussed the history of AI. We have also discussed matrices in detail earlier. In this third article in the series on AI, we will discuss more matrix operations. We will also get familiar with a few more Python libraries useful for developing AI.



Before we proceed any further, let us discuss a few significant terms often associated with AI and machine learning. Artificial neural networks (often called neural networks or NNs) are the core of machine learning and deep learning. But, what are they? By definition, they are computing models inspired by the biological neural networks that form human brains. I won't be including the image of an NN here because the Internet nowadays is

so full of it. But any person interested in AI has definitely seen one or more of those images, with an input layer on the left, one or more hidden layers in the middle, and an output layer on the right of the image. One important thing to understand is that the weights of edges between each layer, shown in the images we mentioned just now, keep on changing with training. This leads to successful machine learning and deep learning applications.

Supervised learning and unsupervised learning are two important models of machine learning. In the long run, any aspirant who wants to work in the field of AI or machine learning needs to learn about these models and the different techniques used to implement them. Hence, I think it is time for us to (informally) understand the difference between these two models. Consider two people named A and B who are supposed to classify apples and oranges into two separate groups. They have never seen apples or oranges. Both are given 100 photos of apples and oranges. But A has the additional information as to which are apples and which are oranges in the given photos. A is like a supervised algorithm. B does not have any additional information. All he/she has is the 100 photos. B is like an unsupervised algorithm. One day both start the classifications. Who will be better at his/her job? Conventional wisdom says A will outperform B. But machine learning says this may not be the case all the time. What if, out of the 100 images given to A, only 5 are of apples and the rest are all of oranges? He/she may not get familiar with apples at all. What if somebody intentionally cheated A by changing some of the labels of the photos, from apples to oranges and vice versa? In these scenarios, B could outperform A in the job.